🚀 GitHub Commands Cheat Sheet

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🟢 1. Pehli baar Git Repository banana (Sirf ek baar)

```
cd Coding

git init

git add .

git commit -m "Initial commit"

git branch -M main

git remote add origin https://github.com/USERNAME/REPOSITORY.git

git push -u origin main
```

Use Case

- Naya project start kiya.
- GitHub se connect karna hai.
- Ye commands **sirf ek baar** chalti hain.

🟢 2. Roz ka Workflow (Most Important ⭐⭐⭐⭐⭐)

```
git add .

git commit -m "Meaningful Commit Message"

git push
```

Use Case

- Nayi file banayi
- Naya folder banaya
- Code edit kiya
- Bug fix kiya
- Project complete kiya

Ye 3 commands tum sabse zyada use karoge.

🟢 3. Repository ki Current State Dekhna

```
git status
```

Use Case

Check karna hai ki

- Kaunsi files change hui
- Kaunsi files add hui
- Kaunsi files delete hui
- Push karna baaki hai ya nahi

🟢 4. Repository Download Karna

```
git clone <Repository URL>
```

Example

```
git clone https://github.com/rohit/coding.git
```

Use Case

- Kisi existing repository ko download karna
- Dusre PC me project lana

🟢 5. Latest Changes Download Karna

```
git pull
```

Use Case

GitHub pe latest changes aaye hain.

Laptop me lana hai.

🟢 6. Sirf Files Stage Karna

```
git add .
```

Use Case

Saare changes stage karna.

Sirf ek file

```
git add main.cpp
```

Sirf ek folder

```
git add React/
```

🟢 7. Commit Banana

```
git commit -m "Completed Binary Search Questions"
```

Use Case

Apne changes ka snapshot save karna.

🟢 8. GitHub Pe Upload Karna

```
git push
```

Use Case

Commit GitHub pe bhejna.

🟢 9. Commit History Dekhna

```
git log
```

Short version

```
git log --oneline
```

Use Case

Purane commits dekhna.

🟢 10. Difference Dekhna

```
git diff
```

Use Case

Commit karne se pehle kya change hua dekhna.

🟢 11. Remote Check Karna

```
git remote -v
```

Use Case

Kaunsa GitHub repository connected hai.

🟢 12. Branch Dekhna

```
git branch
```

Use Case

Current branch check karna.

🟢 13. Branch Banana

```
git checkout -b feature-login
```

Use Case

Naya feature banana.

Abhi tumhe iski zarurat nahi.

🟢 14. Branch Change Karna

```
git checkout main
```

Use Case

Main branch me wapas aana.

🟢 15. Last Commit Ki Details

```
git show
```

Use Case

Last commit me kya hua tha.

🟢 16. Git Configuration

Username

```
git config --global user.name "Rohit Kumar"
```

Email

```
git config --global user.email "yourmail@gmail.com"
```

Use Case

Pehli baar Git install karne ke baad.

🟢 17. Git Version

```
git --version
```

Use Case

Git install hua hai ya nahi.

🟢 18. Current Folder Ka Path

Windows

```
pwd
```

Ya

```
cd
```

Use Case

Abhi kis folder me ho.

🟢 19. Folder Ke Files Dekhna

Windows CMD

```
dir
```

PowerShell

```
ls
```

Git Bash

```
ls
```

🟢 20. Empty Folder GitHub Pe Upload Karna

Git empty folder upload nahi karta.

```
touch FolderName/.gitkeep
```

Windows PowerShell

```
New-Item FolderName\.gitkeep -ItemType File
```

Phir

```
git add .
git commit -m "Added Folder"
git push
```

⭐ Daily Workflow (Ye Yaad Kar Lo)

```
git status

git add .

git commit -m "Your Commit Message"

git push
```

Ye **4 commands** tum roz use karoge.

⭐ Jab Naya Laptop Ya PC Ho

```
git clone <Repository URL>

cd RepositoryName
```

Bas.

⭐ Jab GitHub Se Latest Code Lana Ho

```
git pull
```

⭐ Sabse Important Commands

Command	Kaam

`git status`	Current changes dekhna
`git add .`	Sab changes stage karna
`git commit -m "message"`	Snapshot banana
`git push`	GitHub pe upload karna
`git pull`	GitHub se latest lana
`git clone URL`	Repository download karna
`git log --oneline`	Commit history dekhna
`git diff`	Kya change hua dekhna
`git remote -v`	Connected GitHub repository dekhna
`git branch`	Branch check karna

1. Folder ke andar jaana

```
cd Coding
```

Kya karta hai?

`cd` ka matlab **Change Directory** hota hai.

Ye terminal ko batata hai ki ab `Coding` folder ke andar kaam karna hai.

Example:

```
Desktop/
|
├── Coding/
|    ├── React/
|    ├── C++
|    └── LeetCode/
```

Agar tum Desktop par ho aur `Coding` ke andar jana hai:

```
cd Coding
```

Ab jitni bhi Git commands chalengi, wo `Coding` folder par apply hongi.

---

2. Git Repository banana

```
git init
```

Kya karta hai?

Ye current folder ko **Git Repository** bana deta hai.

Is command ke baad ek hidden folder ban jata hai:

```
Coding/
│
├── .git/
├── React/
├── C++
└── LeetCode/
```

`.git` folder kya hota hai?

Isme Git saari information store karta hai.

Jaise:

- Commit history
- Branches
- Configuration
- Tracking information

Ye folder hi Git ka "brain" hai.

> `git init` sirf ek baar chalana hota hai.

---

3. Saare changes ko Stage karna

```
git add .
```

Kya karta hai?

`.` ka matlab hota hai

> **Current folder aur uske andar ki sab files.**

Ye command bolti hai:

> "Jo bhi naye changes hue hain, unhe commit ke liye ready kar do."

Example:

```
Coding/

React/
    App.jsx

LeetCode/
    TwoSum.cpp

Notes/
    Git.pdf
```

Agar ye sab naye ya modified hain

```
git add .
```

to Git sabko stage kar dega.

---

4. Commit banana

```
git commit -m "Initial commit"
```

Kya karta hai?

Commit matlab

> **Current project ka snapshot.**

Git project ki ek copy save kar leta hai.

Example:

```
Commit 1
↓

React
LeetCode
C++
```

Baad me aur changes karoge

```
Commit 2
↓

React
LeetCode
Projects
```

Is tarah history banti rehti hai.

`-m` kya hai?

`-m` ka matlab

Message

Jo tum likhte ho

```
"Initial commit"
```

wo commit ka naam hota hai.

Examples

```
"Added Login Page"

"Completed Binary Search"

"Fixed Navbar"

"Solved 5 LeetCode Problems"
```

---

5. Main Branch banana

```
git branch -M main
```

Kya karta hai?

Git me code branches me rehta hai.

Sabse important branch hoti hai

```
main
```

Purane Git versions me default branch

```
master
```

hua karti thi.

Ye command current branch ka naam `main` kar deti hai.

---

6. GitHub Repository connect karna

```
git remote add origin https://github.com/USERNAME/REPOSITORY.git
```

Kya karta hai?

Abhi tak Git sirf tumhare laptop par tha.

Is command se GitHub se connection ban jata hai.

Example

```
Laptop
     |
     |
     ▼
Git Repository
     |
     |
     ▼
GitHub Repository
```

`origin` kya hai?

`origin` ek nickname hai.

Iska matlab

```
GitHub Repository
```

Ho gaya.

Actual URL

```
https://github.com/USERNAME/REPOSITORY.git
```

Ko yaad rakhna mushkil hai.

Isliye usko short name de diya

```
origin
```

Ab future me

```
git push origin main
```

likhne se Git samajh jata hai

```
origin = GitHub wala repository
```

---

7. GitHub pe Upload karna

```
git push -u origin main
```

Kya karta hai?

Laptop ke commits ko GitHub par bhej deta hai.

```
Laptop
      |
      | Push
      ▼
GitHub
```

`-u` kya karta hai?

Ye **upstream** set karta hai.

Iska matlab

Git yaad rakhega ki

```
main
```

branch ko

```
origin/main
```

ke saath connect karna hai.

Isliye pehli baar:

```
git push -u origin main
```

Aur uske baad sirf:

```
git push
```

kaafi hota hai.

Multi-Laptop / Multi-OS GitHub Guide

💻 Scenario

Maan lo tere paas:

- 🪟 Laptop 1 → Windows
- 🐧 Laptop 2 → Fedora

Aur GitHub pe ek repository hai:

```
coding-journey
```

🟢 STEP 1: Pehli baar Repository banana (Sirf ek baar)

Ye sirf tab jab repository **local folder se pehli baar GitHub par bhej raha hai**.

```
cd Coding

git init

git add .

git commit -m "Initial commit"

git branch -M main

git remote add origin https://github.com/USERNAME/coding-journey.git

git push -u origin main
```

Iske baad kabhi bhi is repository me `git init` dobara nahi karna.

🟢 STEP 2: Dusre Laptop / Dusre OS par Pehli Baar

Maan lo ab Fedora install kiya.

Sabse pehle Git install aur configure karo.

```
git config --global user.name "Rohit Kumar Rajak"

git config --global user.email "your-email@gmail.com"
```

Ye har naye OS me sirf ek baar.

Ab repository download karo.

```
git clone https://github.com/USERNAME/coding-journey.git
```

Ye command

✅ Naya folder banayegi

```
coding-journey/
```

Aur uske andar

- saara code
- `.git`
- branches
- GitHub connection

sab automatically aa jayega.

Isliye

❌ `git init` nahi

❌ `git remote add origin` nahi

Kuch bhi nahi.

Bas

```
cd coding-journey
```

Aur coding start.

🟢 STEP 3: Dusre Laptop par Coding Karna

Coding karo.

Example

```
React/
    TodoApp
```

Banaya.

Ya

```
LeetCode/
```

me questions solve kiye.

Uske baad

```
git status

git add .

git commit -m "Completed React Todo App"

git push
```

Ho gaya.

GitHub update ho gaya.

🟢 STEP 4: Ab Purane Laptop par Wapas Aana

Windows me abhi purana code hai.

GitHub pe Fedora wala latest code aa chuka hai.

Coding shuru karne se pehle

Repository ke andar jao

```
cd coding-journey
```

Phir

```
git pull
```

Ye latest changes GitHub se isi folder me update kar dega.

❌ Naya folder nahi banega.

Existing folder hi update hoga.

Ab coding karo.

🟢 STEP 5: Windows par Coding Khatam

```
git status

git add .

git commit -m "Solved Tree Problems"
```

```
git push
```

Ho gaya.

GitHub update.

🟢 STEP 6: Fedora par Dubara Jana

Coding shuru karne se pehle

```
git pull
```

Latest code aa gaya.

Coding karo.

```
git add .

git commit -m "Added Binary Search"

git push
```

Bas.
